

AYN THOR · LOWER-SCREEN ASSISTANT



Heimdall Installation & User Guide

AYN Thor lower-screen assistant · Quick start and feature reference

v0.1.1-alpha.1 · English · August 2026



Real-device AYN Thor use

Good to know Installing Heimdall does not require root or Developer options. Developer options are needed only for Shizuku-based controller features.

<https://github.com/mastercook777/Heimdall-AYN-Thor-Assistant/releases>

1. What Heimdall Is

Heimdall keeps the game on Thor's upper screen while the lower screen becomes a persistent control and reference console. Profiles, Macros, touch controls, maps, Guides, static-image Canvases, magnification, recording, and Quick Actions stay within thumb reach.

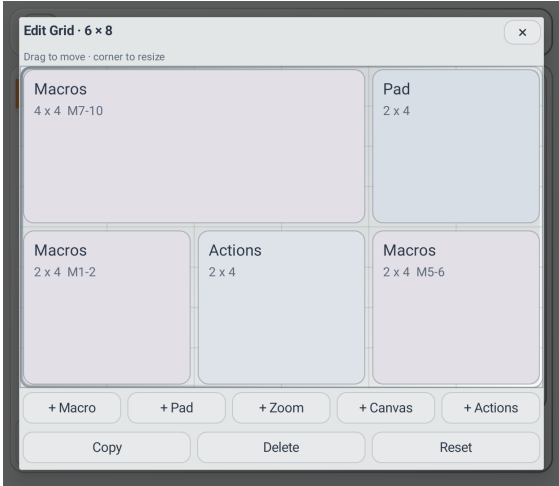
Alpha scope This release targets AYN Thor. Compatibility is not guaranteed for every firmware, controller mode, game, emulator, or other dual-screen device.

Choose only the route you need

What you want to use	Required setup
Profiles, Grid, maps, Guides, and Canvas	Install Heimdall only. Developer options and Shizuku are not required.
Upper-screen taps, holds, swipes, or touchpad drag	Enable the Heimdall Accessibility service for Basic Touch.
Controller Macros, Virtual Right Stick, Precision Aim, Enhanced Touch	Install, start, and authorize Shizuku. Developer options are needed only for wireless-debugging startup.
Live magnifier or recording	Accept Android's screen-capture consent when requested.



Play view: Macros, Quick Actions, and touchpad



Edit view: drag modules and resize from visible corners

2. Download, Install, And Update

- 1 Open the [Heimdall Releases](#) page and select the newest versioned Alpha.
- 2 Download the APK named like heimdall-vX.Y.Z-alpha.N.apk; the same Release also provides SHA256SUMS.txt.
- 3 Open the APK with Thor's file manager. If Android asks, allow that file manager to install unknown Apps, then finish installation.
- 4 Open Heimdall, create one minimal Profile, then enable Accessibility or Shizuku only for the features you need.

Updating from the public Alpha

0.1.1-alpha.1 is designed to update public 0.1.0-alpha.1 in place. Do not uninstall first: uninstalling removes local Profiles and imported assets. Export important Profiles before updating, then recheck Accessibility, Shizuku, and MediaProjection state.

Moving from an older test package

- 1 Export the current or all Profiles from the old App and keep the file outside Heimdall.
- 2 Install the public Alpha. Test packages and the public package may be separate Apps and do not automatically share data.
- 3 Import the file into the public Alpha and verify important Profiles before deleting the old App or backup.

Self-contained Profile bundles

Current builds export .heimdall-profile: configuration plus supported Profile icons, maps, file Guides, user Macro icons, and Canvas images. Import validates paths, sizes, media types, and SHA-256 before replacement. Legacy configuration-only JSON remains importable but cannot restore source assets older builds never bundled.

3. Profiles And Basic Touch

Create your first Profile

- 1 Open Profile management in Settings, create a Profile, and give it a clear name.
- 2 Optionally add a package hint for conservative App-aware matching; Heimdall does not switch when evidence is ambiguous.
- 3 Use the visual 6 × 8 Grid to add, drag, and resize modules. Preview first, then save explicitly.
- 4 Choose touch mode, maps, Guides, and Canvases for the Profile, then test one simple action.

Enable Basic Touch

- 1 Open Android Settings and go to the Accessibility services page.
- 2 Find Heimdall, enable the service, and review Android's confirmation.
- 3 Return to Heimdall and select Basic Touch for the current Profile.
- 4 Test one upper-screen tap or short drag before configuring longer Macros.

Mapper coexistence Basic Accessibility touch and Thor's built-in mapper use separate touch streams and may cancel each other. When coexistence is required, use mapping-compatible Enhanced Touch only when Heimdall reports it available.

Never enabled Developer options?

- 1 Open About device/About handheld in Android Settings and locate Build number.
- 2 Tap Build number repeatedly until Android confirms Developer options are enabled.
- 3 Return to Settings and search for Developer options or Wireless debugging. Menu names may vary by Thor firmware.
- 4 This flow is needed only for Shizuku-based Controller Enhancement.

4. Shizuku And Controller Enhancement

Pair and start Shizuku

- 1 Install Shizuku from the [official Shizuku Releases page](#).
- 2 Choose Start via wireless debugging in Shizuku and begin pairing.
- 3 Return to Android's Wireless debugging page and choose Pair device with pairing code.
- 4 Enter Android's displayed code in the Shizuku notification or pairing screen.
- 5 Return to Shizuku, tap Start, and wait for it to report that the service is running.
- 6 Open Heimdall > Settings > Connection > Controller Enhancement, authorize/connect, and allow Shizuku permission.
- 7 Return to Heimdall and confirm Controller Enhancement is available.

After reboot Check Shizuku before using controller features. If the service stopped, start it again; pairing normally does not need to be repeated.

Enhanced Touch protection

Every Profile keeps Enhanced Touch controller-Macro protection enabled by default. If a Macro contains physical-controller steps, Heimdall rejects the whole Macro before emitting any step to avoid conflict with Thor's mapper. Disable protection only when the built-in mapper is off; screen steps still use the strict mapping-compatible route and never fall back to Accessibility.

If Shizuku stays on "Searching for pairing service," allow background operation and notifications, keep local networking and wireless debugging available, and toggle wireless debugging once. See the [official Shizuku setup guide](#).

5. Grid, Macros, And Play Tools



A local Guide without covering the game



Canvas move list and structured Macros

Edit the Grid

Add, drag, resize, or remove modules in the visual editor. Modules snap to the 6 × 8 Grid; preview first, then save the Grid and Profile explicitly.

Create a touch Macro

Add a Macro module, hold its button for about 1.8 seconds to open the structured editor, then add tap, hold, swipe, or wait steps. Capture coordinates directly on the upper-screen target.

Record a physical-controller Macro

Shizuku and an available Controller Enhancement route are required. Thor's mapper may hide mapped buttons; begin recording, briefly return to Android Home to press the button, then return to Heimdall to review and save the sequence.

Touchpad and aiming

Touch drag converts lower-screen finger movement into an upper-screen drag. Virtual Right Stick uses Controller Enhancement; Precision Aim is for small corrections. Start with low sensitivity and confirm the stick returns to neutral after release.

Magnifier, recording, and Canvas

The live magnifier and recording both occupy MediaProjection and cannot run together. One live magnifier is supported per Profile. Canvas is a Profile-owned static-image module with independent rectangular or circular crop presentation.

6. Backup, Limitations, And Feedback

Back up before uninstalling

After an important Grid or Macro setup, export the current or all Profiles and keep the .heimdall-profile file outside Heimdall. A bundle may contain imported maps, Guides, icons, and Canvas images, so review it before sharing. Android platform backup is disabled for this Alpha; uninstalling removes local data.

Known Alpha limitations

- Mapping-compatible Enhanced Touch is verified only on tested Thor firmware, game, and mapper combinations.
- Basic Accessibility touch and Thor's built-in mapper may cancel each other during simultaneous operation.
- Thor's mapper may hide mapped physical buttons from the Macro recorder.
- Only one live magnifier is supported per Profile, and it cannot share MediaProjection with recording.
- After some lower-screen page transitions, a frozen magnifier may omit its Stop marker or retain the final frame after stopping; use Quick Actions or Settings to stop it.

Common problems

Problem	First response
Shizuku unavailable	Confirm the service is running and Heimdall is authorized; check again after a Thor reboot.
Touchpad does nothing	Check the current Profile mode, Accessibility/Shizuku state, and that the game is on the upper screen.
Mapper and touch interrupt each other	Use available Enhanced Touch when coexistence is required; do not mix it with Basic Touch.
Mapped button is not recorded	Begin recording, briefly return to Android Home, and record the button there.
Magnifier does not start	Stop recording first; both features cannot own screen capture together.
Settings were not retained	Tap Save before leaving the Grid editor or Settings.

Send a useful bug report

Use [GitHub Issues](#) and include the Heimdall version, Thor firmware, native/Xbox controller mode, game or emulator, Basic Touch/Shizuku route, whether Thor mapping is enabled, exact reproduction steps and frequency, and shareable screenshots or recordings.